

Proof of Conjecture 00000000934

An unconditionally convergent series in complex $L^1(\mathbb{R}/\mathbb{Z})$

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Abstract

We construct a series in complex-valued L^1 of the circle that converges unconditionally after every termwise sign choice and admits no rearrangement that converges absolutely in the L^1 norm. The construction uses the Fourier monomials with harmonic coefficients. In fact, arbitrary complex unit phases may replace signs. The accompanying Lean 4 project proves the complete existence assertion with the usual almost-everywhere quotient and norm.

1 Statement and conventions

The English source asserts that there is an unconditionally convergent series in $L^1(\mathbb{T})$ which remains unconditionally convergent after every sign change and admits no absolute rearrangement. The exact English and Chinese source is preserved in `ORIGINAL.md`.

We use the complex scalar convention: $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ has normalized Haar measure μ , with $\mu(\mathbb{T}) = 1$. For this circumference-one circle, μ is also the usual Lebesgue volume. The space $L^1 = L^1(\mathbb{T}; \mathbb{C})$ consists of integrable functions modulo equality almost everywhere, with $\|f\|_1 = \int_{\mathbb{T}} |f| d\mu$. Unconditional convergence means convergence in this norm of the net of sums over all finite subsets of the index set. A sign choice is any function $\varepsilon : \mathbb{N} \rightarrow \{-1, 1\}$, with no support or eventual-constancy restriction. An absolute rearrangement would be a bijection $\pi : \mathbb{N} \rightarrow \mathbb{N}$ for which $\sum_n \|f_{\pi(n)}\|_1 < \infty$. Thus “absolute” refers to the norm of the stated Banach space, rather than an alternative pointwise convergence condition.

The source’s parenthetical reference to the boundary of Orlicz’s theorem is retained as descriptive attribution. It specifies no additional endpoint parameter or sharpness assertion. We make no separate historical or endpoint claim.

Theorem 1. *There exists $f : \mathbb{N} \rightarrow L^1(\mathbb{T}; \mathbb{C})$ such that $\sum_n f_n$ is unconditionally convergent, every infinite termwise scalar sign choice also gives an unconditionally convergent series, and every permutation fails to give an absolutely norm-convergent series.*

2 Construction and unconditional convergence

For $n \geq 1$, let

$$e_n(x) = \exp(2\pi i n x), \quad f_n = \frac{e_n}{n}, \quad f_0 = 0.$$

The exponential is invariant under integer translations of x , so it is well-defined on $\mathbb{T}$. Each term is bounded and integrable. The initial zero term only permits indexing by $\mathbb{N}$; it does not affect any convergence assertion.

The functions e_n are orthonormal in $L^2(\mathbb{T}; \mathbb{C})$:

$$\int_{\mathbb{T}} \overline{e_m} e_n d\mu = \begin{cases} 1, & m = n, \\ 0, & m \neq n. \end{cases}$$

For distinct integers m, n , the integrand is the character e_{n-m} . Translation by $1/(2(n-m))$ multiplies it by -1 , while Haar invariance preserves its integral; the integral is therefore zero. For $m = n$ the integrand is identically one.

Fix an arbitrary sequence $\varepsilon_n \in \mathbb{C}$ with $|\varepsilon_n| = 1$. For finite sets $F, G \subset \mathbb{N}$, orthogonality gives

$$\left\| \sum_{n \in F} \varepsilon_n f_n - \sum_{n \in G} \varepsilon_n f_n \right\|_2^2 = \sum_{\substack{n \in F \Delta G \\ n \geq 1}} \frac{1}{n^2}.$$

Given $\delta > 0$, convergence of $\sum_{n \geq 1} n^{-2}$ gives M for which the tail above M is less than δ^2 . If both F and G contain $\{0, \dots, M\}$, the displayed difference has norm less than δ . The finite-subset net is Cauchy. Completeness of L^2 yields its limit.

For every $g \in L^2$, Cauchy–Schwarz and $\mu(\mathbb{T}) = 1$ imply

$$\|g\|_1 = \int_{\mathbb{T}} |g| \cdot 1 d\mu \leq \left(\int_{\mathbb{T}} |g|^2 d\mu \right)^{1/2} = \|g\|_2.$$

The identity on almost-everywhere classes is consequently a bounded linear map $L^2 \rightarrow L^1$. It sends the convergent finite-subset net to a convergent net in the actual L^1 norm. Hence $\sum_n \varepsilon_n f_n$ converges unconditionally for every unit-phase sequence. The original series follows by taking all phases equal to 1; every requested sign choice is included. The limiting element may depend on the phase sequence.

For any fixed phase sequence, every bijection π gives the same norm limit: its finite prefix sets $\{\pi(0), \dots, \pi(N-1)\}$ eventually contain each prescribed finite subset of $\mathbb{N}$. This also explains directly the relation between the finite-subset and permutation formulations.

3 Failure of every absolute rearrangement

The pointwise equality $|e_n(x)| = 1$ and $\mu(\mathbb{T}) = 1$ give

$$\|f_n\|_1 = \frac{1}{n} \quad (n \geq 1), \quad \|f_0\|_1 = 0.$$

Unit phases preserve these norms. The harmonic series diverges, since each block $2^k \leq n < 2^{k+1}$ contributes at least $1/2$. A bijection of indices does not change the supremum of finite sums of a nonnegative family. Therefore for every unit-phase sequence and every permutation π ,

$$\sum_{n=0}^{\infty} \left\| \varepsilon_{\pi(n)} f_{\pi(n)} \right\|_1 = +\infty.$$

In particular no absolute rearrangement of the original series exists. Together with unconditional convergence above, this proves all clauses of the theorem for a single sequence. $\square$

4 Formal correspondence and verification

The project pins Lean 4.19.0 and Mathlib v4.19.0. All declarations below are in the namespace `Conjecture934` of `Solution.lean`.

Mathematical content	Lean declarations
Actual circle, measure, a.e. quotient and norm	<code>Circle</code> , μ , <code>L1</code> , <code>L2</code> , <code>measure_eq_volume</code> , <code>L1_norm_integral</code>
Explicit Fourier sequence and representatives	<code>term</code> , <code>term_ae</code> , <code>representative_on_real</code>
Bounded identity inclusion $L^2 \rightarrow L^1$	<code>inclusionLinear_exists</code> , <code>inclusionLinear_apply</code> , <code>inclusion_norm_le</code> , <code>inclusion_exists</code> , <code>inclusion_apply</code> , <code>inclusion_fourier</code>
Every infinite phase and sign choice	<code>phase_L2_summable</code> , <code>phase_summable</code> , <code>IsSignChoice</code> , <code>sign_summable</code>
Explicit norm limits and all permutations	<code>phase_unconditional_norm</code> , <code>phase_permutation_norm</code>
Harmonic norm and no absolute ordering	<code>term_norm</code> , <code>no_absolute_rearrangement</code> , <code>phase_norm_partial_sums_diverge</code>
Complete existential source assertion	<code>conjecture</code>
Stronger assertion for every unit phase	<code>strengthened_conjecture</code>

Mathlib’s `Summable` means existence of a limit for the net of finite subset sums. Here it uses the norm topology of `MeasureTheory.Lp`. The proof invokes the library’s orthonormal Fourier family and the orthogonal-family square-summability theorem, then transports convergence through the explicitly established inclusion. Chosen inclusion maps are selected from proved existence statements; pointwise action lemmas fully fix their values on every input. They introduce no additional assumptions.

The frozen mathematical core passes a fresh source replay with warnings treated as errors. Its audit inspects all 36 owned declarations, including generated helpers, and 30,946 transitive declarations. Only the standard axioms `propext`, `Classical.choice`, and `Quot.sound` occur; the audit accepts no unsafe declaration or owned axiom. No numerical computation is needed for the mathematics. The auxiliary programs check provenance and proof dependencies.

`PROOF.md` gives the full declaration mapping; `VERIFICATION.md` retains both successful checks and resolved development failures. The final package’s `VERIFICATION-PACKAGE.md` provides the reviewer workflow and additional verification records. Local compilation and independent review do not constitute maintainer acceptance. This submission claims the complex-valued theorem stated above, without a separate formalized real-valued or Orlicz-endpoint assertion.